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SUPERSEDING
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~~FEDERAL~~ SPECIFICATION

SPRINKLER, LAWN, (SURFACE CONNECTED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers water activated surface connected sprinklers for watering, soaking or sprinkling lawns and gardens.

1.2 Classification.

1.2.1 Types and classes. Sprinklers shall be furnished in the following types and classes as specified (see 6.2):

Type I - Impulse head
- class a - head only
- class b - head with sled base
- class c - head with ring base
- class d - head with wheel base

Type II - Multiple T head

Type III - Rotary head
- class e - head with sled base
- class f - head with spike base
- class g - head with wheel base

Type IV - Oscillating head
- class h - head with wheel base
- class j - head with seld base

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

QQ-Z-363 - Zinc-Base Alloy; Die Castings.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-640 - Boxes, Folding, Fiberboard, Corrugated, Triple Wall.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Set-Up, Paperboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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Military Specification:

MIL-P-116 - Preservation - Packaging, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D256 - Impact Resistance of Plastics and Electrical Insulating Materials.
D638 - Tensile Properties of Plastics.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Certification. Sprinklers offered under this specification shall be certified by the contractor to have successfully completed the endurance test (see 4.4.6 and 6.2).

3.2 Materials. The material shall conform to the requirements specified hereinafter. Materials not definitely specified shall be of the quality normally used for the purpose in commercial practice.

3.2.1 Heads and bases. Unless otherwise specified (see 6.2), the heads and bases shall be constructed of corrosion resistant metal such as aluminum, brass, bronze, zinc, nickel-copper alloy, stainless steel, or zinc or other suitable protective coating on steel and shall be of sufficient thickness to assure adequate strength.

3.2.1.1 Class b sled base. The base may be fabricated from a suitable plastic, reinforced if necessary, provided the plastic base meets all other applicable requirements.

3.2.1.2 Nickel copper alloy. Commercial nickel copper alloy shall contain not less than 63 percent nickel.

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3.2.1.3 Stainless steel. This alloy shall contain not less than 11.5 percent chromium.

3.2.1.4 Zinc-base-alloy die castings. Die castings of zinc-base-alloys may be used provided they conform to QQ-Z-363 as to material, workmanship, manufacture, and chemical composition. Either composition specified therein may be furnished.

3.2.1.5 Plastic. Unless otherwise specified (see 6.2) plastic for components shall be acetal, acrylic, fluoro-carbon, nylon, polypropylene, acrylonitrite butadiene styrene or polystyrene.

3.3 Gears and bearings. Gears and bearings shall be constructed of copper alloy or any suitable plastic covered by 3.2.1.5.

3.4 Finish. All exposed or visible metal parts shall be enameled, except for corrosion resisting steel, copper alloy, and aluminum. Zinc base alloy die castings shall be cadmium, chromium, zinc plated or have an enameled painted finish of any color, at the option of the manufacturer.

3.5 Design.

3.5.1 Type I, impulse head. The impulse sprinkler head shall be designed to throw water in a full or partial circle. The sprinkler shall be equipped with an adjustable break-up device to regulate radius of coverage, spring loaded to provide positive setting. The sprinkler housing shall have 1/2 inch or 3/4 inch external American Standard taper pipe threads (NPT) as specified (see 6.2), in accordance with National Bureau of Standards (NBS) Handbook H-28 "Screw thread Standards for Federal Services."

3.5.2 Type II, Multiple T head. The multiple T type sprinkler shall be designed to spray water in a round or square pattern. The sprinkler shall be provided with a flow-thru 3/4 inch metal hose connection, with male and female connections, so that the sprinkler may be used in a series of two or more, or as a single spray. The base shall be rigid and permanently affixed and shall be of sufficient length and width to maintain stability when in use. The valve in the head shall be adjustable to control the spray. Each sprinkler shall be provided with an end cap for the male connection.

3.5.3 Type III, Rotary head. The rotary type head shall be designed to spray water by a whirling action in a square or circular pattern, as specified (see 6.2). The sprinkler shall be either 1, 2, or 3 arm design, as specified (see 6.2), and shall be equipped with or without a mechanical adjustment to make the sprinkler either stationary or rotary, as specified (see 6.2). The nozzles on the arm ends shall be fixed or adjustable nozzles, the threads on the outer ends of the arms shall have sufficient wall thickness remaining to withstand forces exerted when the sprinkler is in use.

3.5.4 Type IV, Oscillating head. The oscillating type sprinkler shall be designed to spray water in a backward and forward motion over a rectangular pattern. The sprinkler shall be equipped with a control knob for adjusting the spray to center-to-right, center-to-left, partial or full sweeps. When specified (see 6.2), the sprinkler shall be constructed with an automatic timing device to shut off flow of water. The sprinkler shall be constructed with a corrosion resistant metal spray bar.

3.5.5 Classes b and e, sled type base. The sled base shall be of sufficient length and width to provide stability when attached to a hose and pulled across a level turf and also maintain the sprinkler in a stable upright position when sprinkler is in use. The sled base shall be constructed with a 1/2 inch or 3/4 inch sprinkler connection, as specified (see 6.2), and a 3/4 inch plastic or corrosion resisting metal hose connection with an elastomer washer.

3.5.6 Class j, sled base. The design of the sled base for type IV head shall be at the option of the manufacturer. The sled base shall be of sufficient length and width to provide stability when attached to a hose and pulled across a level turf and also to maintain the sprinkler in a stable upright position when in use. This base in an integral part of the oscillating sprinkler and is not adaptable for use with types I, II, and III heads. This base shall be equipped with a 3/4 inch corrosive resistant metal swivel hose connection, with elastomer washer.

3.5.7 Class c, ring base. The ring base shall be formed from rigid metal of an adequate configuration to provide sprinkler stability when watering large areas. The ring base shall be provided with a 1/2 inch or 3/4 inch sprinkler connection as specified (see 6.2) and a 3/4 inch plastic or corrosion resisting metal swivel hose connection, with elastomer washer.

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3.5.8 Classes d and g, wheel base. The wheel base shall be of molded construction, with 3 rigid arms forming the base, one arm of which will act as a rest and house the water connection. The base shall have a minimum of 2 wheels, large enough to permit easy dragging across the turf. The wheel base shall be of sufficient length and width to maintain sprinkler balance. A 3/4 inch corrosive resistant metal swivel hose connection with elastomer washer shall be provided.

3.5.9 Class h, wheel base. The design of the wheel base for type IV head shall be at the option of the manufacturer. This base forms an actual part of the oscillating sprinkler and is not adaptable for use with types I, II, and III heads. A 3/4 inch corrosive resistant metal swivel hose connection, with elastomer washer shall be provided.

3.5.10 Class f, spike base. The spike base shall be constructed with a pointed end to permit easy entry into the turf and shall be rigid enough to maintain stability during sprinkler operation. The spike base shall be provided with a 1/2 inch sprinkler connection and a 3/4 inch corrosive resistant metal swivel hose connection with elastomer washer.

3.5.11 Type III base. Single or dual pipe supply shall be provided, as specified (see 6.2). When dual pipe supply is required an end cap for male connection shall be provided.

3.6 Performance.

3.6.1 Type I, Impulse head. The sprinkler equipped with a 1/2 inch connection shall be capable of providing a uniform distribution of water over a circular area of 75 ± 5 feet in diameter at a water pressure of 40 ± 5 psi at the sprinkler for a period of at least 10 minutes. The sprinkler equipped with a 3/4 inch connection shall distribute water in a circular pattern of 90 ± 5 feet in diameter at a water pressure of 50 ± 5 psi at the sprinkler for a period of at least 10 minutes.

3.6.2 Type II, Multiple T head. The sprinkler shall be capable of spraying water in a circular area of 20 ± 5 feet in diameter or a square pattern of 20 ± 5 feet at a water pressure of 20 ± 2 psi at the sprinkler for a period of at least 10 minutes.

3.6.3 Type III, Rotary head. The sprinkler shall be capable of spraying water over an area 40 ± 5 feet square, or over a circular area 40 ± 5 feet in diameter at a water pressure of 40 ± 5 psi at the sprinkler for a period of at least 10 minutes.

3.6.4 Type IV, Oscillating head. The sprinkler shall be capable of providing uniform distribution of water over a rectangular area of 2,000 square feet or greater as specified (see 6.2) at a water pressure of 40 ± 5 psi at the sprinkler for a period of at least 10 minutes.

3.6.5 Endurance. Types I, II, III, and IV heads shall not leak in a steady stream or become disengaged when tested in accordance with 4.4.6.

3.6.6 Plastic. Plastics used for the class b sled base shall meet the following requirements.

3.6.6.1 Impact strength. When tested in accordance with 4.4.7, the impact strength of plastics used for class b sled bases shall be a minimum of 0.45 ft-lbs / inch of notch.

3.6.6.2 Tensile strength. When tested in accordance with 4.4.8, the tensile strength of plastics used for class b sled bases shall be minimum of 4000 psi.

3.7 Identification marking. The sprinkler shall be marked in a plain or permanent manner with the manufacturer's name or trade mark of such known character that the source of manufacturer may be readily determined.

3.8 Workmanship. Sprinklers shall be free from sharp edges, projections and defects which might impair its serviceability or appearance. Surfaces shall be free from flaws and cracked or peeling plating or paint.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the preformance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of each type and class of sprinklers submitted for inspection at the same time.

4.2.2 Visual, dimensional, and workmanship of end item. Sprinklers shall be examined for these characteristics in accordance with level S-3 in MIL-STD-105.

4.2.3 Sampling for tests. Samples for tests shall be in accordance with level S-2.

4.3 Inspection.

4.3.1 End item. Sprinklers shall be examined for defects listed in table I. The Acceptable Quality Levels (AQL's) for defects listed in table I shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

TABLE I. Classification of defects, end item

Defects	Major	Minor
Type and class not as specified	X	
Sharp edges and projections	X	
Leakage at joints, bearings, and connections	X	
Parts disengaged	X	
Metal surfaces not completely painted or plated	X	
Sled base sprinkler turns over when dragged across level turf	X	
Sprinkler unstable when in operation	X	
Plated or painted surfaces, loose or scratched		X
Damage or defects not affecting function or serviceability		X

4.3.2 Inspection of preparation for delivery requirements. An inspection shall be made to determine whether the packaging and marking requirements comply with section 5 of this specification. Defects shall be scored in accordance with table II. For examination of interior packages the sample unit shall be one shipping container fully prepared for delivery selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE II. Class of preparation for delivery defects

Examine	Defects
Markings	Omitted; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling.
	Bulging or distortion of container.

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4.4 Tests.

4.4.1 Acceptable Quality Level. The AQL for tests shall be 6.5 percent.

4.4.2 Water pressure. Water pressure shall be regulated in accordance with the applicable requirements of 3.6.1, 3.6.2, 3.6.3, and 3.6.4.

4.4.3 Sprinkling area. Tests shall be conducted with the sprinkler mounted at ground level on a surface such as concrete or black top, so that water coverage may be readily measured (see 3.6.1, 3.6.2, 3.6.3, and 3.6.4).

4.4.4 Performance. The sprinklers shall be operated at prescribed water pressure for a period of 10 ± 2 minutes (see 3.6.1, 3.6.2, 3.6.3, and 3.6.4).

4.4.5 Endurance. Types I, III, and IV sprinklers shall be operated at a water pressure of 40 + 5 psi at the sprinkler, and type II sprinkler at a water pressure of 20 ± 2 psi at the sprinkler for a period of 500 hours continuously (see 3.6.5).

4.4.6 Impact strength. Class b bases, if made of plastic, shall be tested for impact strength in accordance with Method A of ASTM D256, using a 0.125 inch specimen (see 3.6.6.1).

4.4.7 Tensile strength. Class b bases, if made of plastic, shall be tested for tensile strength at yield in accordance with ASTM D638, using a 0.125 inch type III specimen at a test speed of 2 inches / minute.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified, (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging.

5.1.1.1.1 Heads only. Each head shall be packaged in accordance with method III of MIL-P-116.

5.1.1.1.2 Sprinklers. Each head with base comprising one complete sprinkler shall be packaged in accordance with method IC-2 of MIL-P-116.

5.1.1.2 Intermediate packaging. Six impulse heads of like description or 10 multiple heads of like description shall be intermediate packaged in a close-fitting box conforming to PPP-B-636, W5c. The box shall be closed in accordance with appendix to the box specification.

5.1.2 Level B.

5.1.2.1 Unit packaging.

5.1.2.1.1 Heads only. Each head shall be packaged as specified in 5.1.1.1.1.

5.1.2.1.2 Sprinklers. Each head with base comprising one complete sprinkler shall be packaged in a close fitting box conforming to PPP-B-566, PPP-B-665, or PPP-B-676. The boxes shall be secured to prevent accidental opening during shipment and storage.

5.1.2.2 Intermediate packaging. Six impulse heads of like description or 10 multiple heads of like description shall be intermediate packaged in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to the box specification.

5.1.3 Level C. The sprinklers shall be packaged to insure protection during shipment and safe delivery at destination.

5.2 Packing. Packing shall be level A, B, or C as specified, (see 6.2).

5.2.1 Level A. Sprinklers of like description, packaged as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-585, class 3, PPP-B-591, class II, PPP-B-601, overseas type, PPP-B-621, class 2, PPP-B-636, class weather resistant, class 2 or PPP-B-640, class 2. The fiberboard boxes shall be closed in accordance with

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the appendix to the box specification. The gross weight of the triple-wall fiberboard box and the wood boxes shall not exceed approximately 200 pounds. The gross weight of the PPP-B-636 fiberboard box shall not exceed the weight limitations of the box specification. Strapping shall be in accordance with the appendix to the applicable box specification.

5.2.2 Level B.

5.2.2.1 Heads only. Twelve intermediate packages of impulse heads or 6 intermediate packages of multiple heads, packaged as specified in 5.1, shall be packed in a box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with method II of the appendix to the box specification.

5.2.2.2 Sprinklers. Unit packages of like description, packaged as specified in 5.1, shall be packed in a close-fitting box conforming to PPP-B-636, class-domestic. The box shall be closed in accordance with the appendix to the box specification.

5.2.3 Level C. The sprinklers, packaged as specified in 5.1, shall be packed to comply with the National Freight Classification rules or the Uniform Freight Classification rules.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2 as specified, (see 6.2).

5.3.1 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Lawn sprinklers are intended to provide a rain like drenching to lawns and gardens, distributing water evenly, without flooding or puddling or excessive errosive action.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class (see 1.2.1).
(c) Whether a test report on certification is required for the endurance test (see 3.1).
(d) Material for heads and bases (see 3.2.1).
(e) Type of plastic (see 3.2.1.5).
(f) Type I, pipe size (see 3.5.1).
(g) Type III - number of arms, whether mechanical adjustment is required (see 3.5.3).
whether square or circular pattern is required (see 3.5.3)
whether single or dual pipe supply is required (see 3.5.11)
whether nozzles shall be fixed or adjustable (see 3.5.3).
(h) Type IV, whether timing device is required (see 3.5.4).
(i) Class b and e, sled type base, pipe size (see 3.5.5).
(j) Class c, ring base, pipe size (see 3.5.7).
(k) Type IV area of coverage (see 3.6.4).
(l) Size of lot is different from 4.2.1.
(m) Level of packaging and packing required (see 5.1 and 5.2).
(n) Whether civil or military markings is required (see 5.3).

6.3 Transportation description. The transportation description and minimum weights applicable to this commodity are: (to be furnished).

Preparing activity: GSA - FSS

Civil Agency Coordinating Activity:

HEW - FEC

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